

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12413913
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Statom; Drew et al.

Method and system for transducer validation

Abstract

A device, a method, and a non-transitory storage medium are described in which a transducer validation service is provided. The service may include stimulating a piezoelectric transducer. The service may include measuring a signal output, which is responsive to the stimulus, from the piezoelectric transducer. The service may include evaluating the measured signal output to a reference voltage and determining whether the piezoelectric transducer is reverse-poled or not based on the evaluation. The service may further include determining a response to the determination when the piezoelectric transducer is reverse-poled.

Inventors:	Statom; Drew (Sheffield, AL), Matam; Mahesh (Cumming, GA)
Applicant:	Neptune Technology Group Inc. (Tallassee, AL)
Family ID:	85025205
Assignee:	Neptune Technology Group LLC (Tallassee, AL)
Appl. No.:	17/813500
Filed:	July 19, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230036761 A1	Feb. 02, 2023

Related U.S. Application Data

us-provisional-application US 63227457 20210730

Publication Classification

Int. Cl.: H04R17/00 (20060101); G01F1/66 (20220101); G01F1/667 (20220101); G01H3/00 (20060101); G01H13/00 (20060101); G01N29/30 (20060101); G01S7/52 (20060101)

U.S. Cl.:

CPC H04R17/005 (20130101); G01F1/662 (20130101); G01F1/667 (20130101); G01H3/005 (20130101); G01H13/00 (20130101); G01N29/30 (20130101); G01N2291/106 (20130101); G01S7/5205 (20130101)

Field of Classification Search

CPC: H04R (17/005); G01F (1/662); G01F (1/667); G01N (29/30); G01N (2291/106); G01H (3/005); G01H (13/00); G01S (7/5205)

USPC: 73/1.82

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3146141	12/1963	Woodland	228/1.1	B29C 66/73921
3391571	12/1967	Johanson	310/368	G01H 1/00
3586888	12/1970	Dorfman	335/205	F23Q 3/002
3648279	12/1971	Watson	310/366	H03K 17/967
3792467	12/1973	Sauveur	310/339	H03K 17/964
3942381	12/1975	Brown	73/703	G01L 11/04
3955409	12/1975	Moser	73/847	G01N 3/32
3963966	12/1975	Mohr	102/210	F23Q 3/002
4015152	12/1976	Mohr	310/339	F23Q 3/002
4485324	12/1983	Berlincourt	310/339	H02N 2/183
4744423	12/1987	Chia-Hsi	173/90	B25D 11/00
4975616	12/1989	Park	310/366	H03K 17/9643
5048320	12/1990	Mitsuhashi	73/12.09	G01M 7/08
5286199	12/1993	Kipke	434/114	H02N 2/043
5325339	12/1993	Yost	367/13	G01N 29/30
5345514	12/1993	Mahdavi	382/152	B82Y 15/00
5431054	12/1994	Reeves	73/612	G01N 29/043
5686675	12/1996	Barton	73/866.5	G01B 17/02
6327921	12/2000	Hsu	73/866.5	G01N 29/0645
7345407	12/2007	Tanner	310/339	H02N 2/188
7780597	12/2009	Panda	310/318	G01S 15/8909
8159900	12/2011	Moore	367/13	G01S 7/52052
8164976	12/2011	Moore	367/13	G01S 7/52052

10168301	12/2018	Böhm	N/A	G01N 29/043
2009/0093999	12/2008	Kearns	702/184	G01N 29/4481
2009/0303064	12/2008	LaBreck	340/678	G01N 29/07
2011/0032799	12/2010	Moore	367/95	G01S 7/52052
2016/0091466	12/2015	Ginther	73/1.82	G01M 99/008
2016/0131746	12/2015	Beaty	367/13	G01S 15/8915
2016/0209371	12/2015	Böhm	N/A	G01N 29/069
2019/0064014	12/2018	Sun	N/A	G01N 29/4472
2019/0242858	12/2018	Yamamoto	N/A	G01N 29/4463
2021/0364471	12/2020	Vrana	N/A	G01N 29/043
2022/0079566	12/2021	Makita	N/A	G16H 40/40
2022/0397500	12/2021	Kim	N/A	G01N 29/4472
2023/0036761	12/2022	Statom	N/A	G01F 1/662

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103436685	12/2012	CN	N/A
103595295	12/2013	CN	N/A
104807890	12/2014	CN	N/A
204594938	12/2014	CN	N/A
106198759	12/2015	CN	N/A
106770669	12/2016	CN	N/A
206876643	12/2017	CN	N/A
112462353	12/2020	CN	G01S 7/52004
102013211616	12/2013	DE	G01N 29/043
3524974	12/2018	EP	G01H 5/00
1150483	12/1985	IT	B41J 5/08
5166724	12/2012	JP	G01N 29/0618
2019082460	12/2018	JP	N/A
20200095642	12/2019	KR	G01M 7/08
1782026	12/2018	TW	A61C 19/04
I782026	12/2018	TW	A61C 19/04
WO-2019119309	12/2018	WO	N/A

OTHER PUBLICATIONS

Diederik Van Nuffel et al., Calibration of Dynamic Piezoelectric Force Transducers Using the Hopkinson Bar Technique, 15th International Conference on Experimental Mechanics, Porto/Portugal, Jul. 22-27, 2012, pg(s) 13 (Year: 2012). cited by examiner

Lianghui Ding et al., Modeling and Evaluation of Piezoelectric Transducer (PZT)-Based Through-

Metal Energy and Data Transfer, Sensors 2020, 20, 3304; doi:10.3390/s20113304, p. 16 (Year: 2020). cited by examiner

Jongseon Johnson et al., An impedance measurement system for piezoelectric array element transducers, Measurement 97 (2017) 138-144 (Year: 2017). cited by examiner

B. Yan et al., Measurement of Strain and Polarization in Piezoelectric and Electrostrictive Actuators, Smart Materials and Structures, Oct. 10-11, 2002, Montreal, Quebec, Canada, p. 7 (Year: 2002). cited by examiner

J.J. Peters et al., A prototype tap test imaging system: Initial field test results, <https://doi.org/10.1063/1.1291323>, AIP Conf. Proc. 509, 2053-2060 (2000) (Year: 2000). cited by examiner

Adams RD et al., Citation: Low-Velocity Impact Inspection of Bonded Structures, Structural Adhesives in Engineering, Bristol; UK; Jul. 2-4, 1986, 1986, p. 139-142 (Year: 1986). cited by examiner

Adams, R D et al., Citation: Review of progress in quantitative nondestructive evaluation. Volume 5B—Proceedings of the Twelfth Annual Review, Williamsburg, VA; Jun. 23-28, 1985, 1985, p. 1253-1258 (Year: 1985). cited by examiner

Diederik Van Nuffel et al., Calibration of Dynamic Piezoelectric Force Transducers Using the Hopkinson Bar Technique, 15th International Conference on Experimental Mechanics, Porto/Portugal, Jul. 22-27, 2012, p. 13 (Year: 2012). cited by examiner

Stewart Sheritt et al., Characterization of Piezoelectric Materials for Transducers, Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California, USA, Nov. 2007, p. 45 (Year: 2007). cited by examiner

J.-H. Tong et al., Fabrication of a Piezoelectric Impact Hammer and Its Application to the In-situ Nondestructive Evaluation of Concrete, Jpn. J. Appl. Phys. vol. 41 (2002) pp. 6595-6600 (Year: 2002). cited by examiner

Tuğrul Öztürk et al, A contribution to the dynamics of the tapping process: analytically estimated and measured instantaneous eigenfrequencies of the tapping tool, Production Engineering (2023) 17:613-624 <https://doi.org/10.1007/s11740-022-01158-3>, p. 12 (Year: 2023). cited by examiner

Yinghua Hu et al., Push-Pull Inverter Using Amplitude Control and Frequency Tracking for Piezoelectric Transducers, Micromachines 2023, 14, 2147. <https://doi.org/10.3390/mi14122147>, p. 18 (Year: 2023). cited by examiner

Prof. DSc. Dževad Hadžihafizović et al, Introduction to Nondestructive Testing NDT, University of Sarajevo 2024, p. 245 (Year: 2024). cited by examiner

Robert G. Bryant, Overview of NASA Langley's Piezoelectric Ceramic Packaging Technology and Applications, NASA Langley Research Center, Hampton VA, 2007, p. 8 (Year: 2007). cited by examiner

Steve Carter et al, Selecting Piezoresistive vs. Piezoelectric Pressure Transducers, Kulite Semiconductor Products, Inc.-Industry/ Kulite General Overview , 2018, pp. 1-25 (Year: 2018). cited by examiner

Lianghui Ding et al., Modeling and Evaluation of Piezoelectric Transducer (PZT)-Based Through-Metal Energy and Data Transfer, Sensors 2020, 20, 3304; doi:10.3390/s20113304, pp. 16 (Year: 2020). cited by examiner

J.W. Waanders, Piezoelectric Ceramics: Properties and Applications, Philips Components, 1991, 91 pages (Year: 1991). cited by examiner

João C. S. Queiroz et al., Damage Detection in Composite Materials Using Tap Test Technique and Neural Networks, Journal of Nondestructive Evaluation (Feb. 22, 2021) 40:27 <https://doi.org/10.1007/s10921-021-00759-9>, p. 9 (Year: 2021). cited by examiner

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to U.S. Provisional Patent Application No. 63/227,457 filed on Jul. 30, 2021, the entirety of which is hereby incorporated by reference herein.

BACKGROUND

(1) Utility companies and other entities operate distribution systems for various resources (e.g., water, gas, electricity, chemicals, etc.) to deliver these resources to customers connected to the distribution systems. A meter may be used at each point the resource is removed and/or provided from the distribution system to a customer to measure usage. For flow metering, the meter may include transducers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating a view of an exemplary apparatus in which an exemplary embodiment of transducer validation may be implemented;
- (2) FIG. 2A is a diagram illustrating another view of a portion of an exemplary apparatus in which an exemplary embodiment of transducer validation may be implemented;
- (3) FIG. 2B is a diagram illustrating yet another view of a portion of an exemplary apparatus in which an exemplary embodiment of transducer validation may be implemented;
- (4) FIG. 3 is a diagram illustrating a view of an exemplary apparatus in which an exemplary embodiment of transducer validation may be implemented;
- (5) FIG. 4A is a diagram illustrating exemplary signals caused by excitation of a transducer;
- (6) FIG. 4B is another diagram illustrating exemplary signals caused by excitation of a transducer;
- (7) FIGS. 5A and 5B are diagrams illustrating views of exemplary embodiments of a transducer holder of an exemplary apparatus;
- (8) FIG. 6 is a diagram illustrating another view of a portion of an exemplary apparatus in which an exemplary embodiment of transducer validation may be implemented;
- (9) FIG. 7 is a diagram illustrating exemplary components of a device may correspond to a device illustrated and described herein;
- (10) FIG. 8 is a flow diagram illustrating an exemplary process of an exemplary embodiment of transducer validation; and
- (11) FIG. 9 is a flow diagram illustrating another exemplary process of an exemplary embodiment of transducer validation.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(12) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements. Also, the following detailed description does not limit the invention.

(13) Checking the polarity of piezoelectric transducers is an important aspect of qualifying transducers for further use in flow metering. For example, two transducers that have opposite polarity when used in a meter, such as a flow meter, may produce artifacts that will affect a flow measurement and its accuracy. A reverse poled transducer may produce a lower pulse width ratio relative to a correctly poled transducer. Additionally, a reverse poled transducer may suffer from

low pulse width ratio. Transducers that are differently poled may produce waveforms that are out of phase with each other (e.g., 180 degrees out of phase).

(14) To avoid using an incorrectly configured transducer, a test may be performed on the transducer. For example, a non-destructive method for testing the transducer may involve the use of x-ray. However, this approach can be problematic because of the gray-scale image and inability to pinpoint the different colored wires that connect to the electrodes. Further, x-ray analysis is not able to determine transducer poling issues (e.g., reversed poled). Alternatively, there are destructive methods for testing, such as cutting open the transducer. Manufacturers or other types of entities may discard transducers that have reverse polarity issues even when using non-destructive methods.

(15) According to exemplary embodiments, an apparatus and method for testing transducers is described. According to an exemplary embodiment, the test may check for oppositely poled transducers. According to an exemplary embodiment, the test may check for reverse-connected wires. According to an exemplary embodiment, the test may be applied to a piezoelectric transducer.

(16) According to an exemplary embodiment, the apparatus may include a frame. The frame may be configured to support a first piece and a second piece. According to an exemplary embodiment, the first piece may be positioned above the second piece. According to an exemplary embodiment, the first piece may be fixed to the frame and the second piece may be removable from the frame. According to an exemplary embodiment, the second piece may be configured to receive a transducer under test. The first piece may include an aperture configured to allow a striker to pass through the aperture and hit a portion of a transducer under test. The second piece may be configured to expose the portion of the transducer and allow the striker to hit the transducer under test. Based on the hitting of the transducer, the transducer may produce a charge. The induced charge or voltage may be measured and evaluated. Based on the result of the measurement and evaluation, a reverse poled transducer and/or oppositely connected wires of the transducer may be detected in a non-destructive manner. Additionally, as described herein, a piezoelectric transducer that may have a polarity issue may not be discarded. For example, as described herein, the logic of a water meter or another type of device may be modified in a manner that accounts for the polarity issue.

(17) FIG. 1 is a diagram of an exemplary apparatus that provides an exemplary embodiment of transducer validation, as described herein. According to various exemplary embodiments, apparatus **100** or portions thereof may be made of metal (e.g., stainless steel, aluminum, tungsten, or the like), plastic (e.g., heavy plastic, polycarbonate (PC), or the like), or another type of material or composite that may have a suitable density, rigidity, and/or other desired properties of relevance.

(18) As illustrated, apparatus **100** may include a frame **103**. Frame **103** may include a base panel **104** and panels **105-1** and **105-2** (also referred to collectively as panels **105** and individually or generally as panel **105**). Panels **105** may extend substantially perpendicular from base panel **104**. Panels **105** may be substantially parallel to each other. According to various exemplary embodiments, base panel **104** and panels **105** may or may not be a unitary piece or structure. Panels **105** may provide stability for frame **103**. Panels **105** may be configured to receive a guide piece **107** and a transducer holder **109**. Panels **105** may also support guide piece **107** and transducer holder **109**, as described herein. According to an exemplary embodiment, guide piece **107** may be affixed to panels **105**. For example, guide piece **107** may be affixed to panels **105** using fasteners **112**, such as screws, bolts, or another type of element that may attach guide piece **107** to panels **105**. The number and placement of fasteners **112** are exemplary. Although not illustrated, according to an exemplary embodiment, transducer holder **109** may be affixed to panels **105** in a similar manner. According to other exemplary embodiments, guide piece **107** may be affixed to panels **105** and transducer holder **109** may not be affixed to panels **105**, as described herein.

(19) Guide piece **107** may align and guide a striker (not illustrated) to hit a transducer (also not

illustrated) that is positioned in transducer holder **109**. For example, guide piece **107** may include a passthrough hole **108** that allows the striker to hit the transducer seated in transducer holder **109** via guide piece **107**, as described herein. For example, a center point of passthrough hole **108** may be vertically aligned with a center point of a passthrough hole **110** formed in a body of transducer holder **109**. According to some exemplary embodiments, passthrough hole **108** and passthrough hole **110** may be concentric in which the diameter of passthrough hole **108** may be smaller than a first bore of passthrough hole **110**, as described herein. Guide piece **107** also may include a face plate **142**, which is described further below. Additionally, for example, transducer holder **109** may include passthrough hole **110** that is configured to receive and secure the transducer under test, as described herein. Transducer holder **109** may also include face plates **143-1** and **143-2** (also referred to collectively as face plates **143** and individually or generally as face plate **143**), which is described further below.

(20) As further illustrated, according to some exemplary embodiments, apparatus **100** may include a mounting piece **115**. Mounting piece **115** may be affixed to frame **103** using fasteners **117**. The number and placement of fasteners **117** are exemplary. According to other exemplary embodiments, frame **103** and mounting piece **115** may be a unitary piece. Mounting piece **115** may be configured to mount a mechanism that automates management of various aspects of the striker during a test, such as positioning and releasing the striker, which causes the striker to hit the transducer under test via guide piece **107** and retrieving the striker for a subsequent test, as described herein. According to other exemplary embodiments, mounting piece **115** may be omitted. For example, an operator of apparatus may manage various aspects of the striker to allow transducers to be tested according to an exemplary embodiment of transducer validation, as described herein. As an example, a human may position and release the striker via passthrough hole **108** and retrieve the striker after the transducer under test has been hit. This process may be repeated.

(21) Understandably, however, human intervention in relation to this aspect of testing the transducer (and perhaps other aspects of a validation test) may not yield optimal results given various factors relating to repeatability (e.g., height and/or position of striker relative to the transducer under test upon release, steadiness of striker upon release, etc.), throughput (e.g., the number of tests performed over time), and/or other types of metrics of relevance. According to still other exemplary embodiments, the mechanism may be free-standing and not attached to apparatus **100**, in which case mounting piece **115** may be omitted.

(22) Although FIG. **1** illustrates and describes an exemplary embodiment of apparatus **100**, according to other exemplary embodiments, apparatus **100** may include additional, different, and/or fewer elements. For example, guide piece **107** and the striker may be omitted. By way of further example, the transducer under test may be stimulated by way of other types of mechanisms and/or methods, such as use of an ultrasonic pulse, a pressure pulse, or another manner that provides a suitable excitation that would cause the piezoelectric transducer to produce a signal, which may be measured and analyzed, to determine whether the transducer is correctly poled or not, as described herein.

(23) FIG. **2A** is a diagram illustrating another view of a portion of apparatus **100**. As illustrated, according to an exemplary embodiment, panel **105** may include a track **120-1** and a track **120-2** (also referred to collectively as tracks **120** and individually or generally as track **120**). For example, track **120-1** may be configured to receive guide piece **107** (not illustrated in FIG. **2A**) and tracking **120-2** may be configured to receive transducer holder **109**. According to this example, each track **120** may have a squared-C configuration. For example, track **120-1** may define the squared-C configuration based on an upper lip **123-1** and a lower lip **125-1**, and track **120-2** may define the squared-C configuration based on an upper lip **123-2** and a lower lip **125-2**. For description purposes, upper lips **123-1** and **123-2** may be referred to collectively as upper lips **123** and individually or generally as upper lip **123**, and similar nomenclature for lower lips **125** or individually or generally as lower lip **125**. As such, upper lips **123** and lower lips **125** of each panel

105 may stabilize and secure guide piece **107** and transducer holder **109** when inserted in tracks **120**. For example, respective portions (e.g., cooperating surfaces) of top and bottom surfaces of guide piece **107** and transducer holder **109** may be in supportive contact with upper lips **123** and lower lips **125**. According to other examples, tracks **120** may be of a shape and/or a dimension different from the squared-C configuration, as described herein. According to some exemplary embodiments, tracks **120** may position guide piece **107** and transducer holder **109** parallel to each other. According to other exemplary embodiments, tracks **120** may not run parallel to each other. For example, referring to FIG. 2B, transducer holder **109** may be slightly angled towards base panel **104**. For example, line **148** may indicate a position for the bottom or second side of transducer holder **109** if it were parallel to guide piece **107**. Here, an angle **149** is illustrated. For example, the range for angle **149** may be about less than 2 degrees but not 0 degrees or substantially 0 degrees. Higher angles (e.g., above this range) may prevent sufficient transfer of stimulus to the piezoelectric material of the transducer.

(24) Referring back to FIG. 2A, as further illustrated between tracks **120**, panel **105** may include spacing portions **127-1**, **127-2**, and **127-3** (also referred to collectively as spacing portions **127** and individually or generally as spacing portion **127**). Spacing portions **127-1** and **127-2** may contribute to a distance that the striker travels to hit the transducer under test with sufficient force to induce a charge or voltage for measurement and analysis, as described herein. Spacing portion **127-2** may afford sufficient space between a bottom surface of guide piece **107** and a top or first surface of transducer holder **109** to allow transducers to be received and removed or swapped in and out for testing. According to various exemplary implementations, transducers may be received and removed or swapped in and out for testing by a human or use of a device. According to another approach, referring to FIG. 1, transducer holder **109** may be slid away from frame **103** (e.g., base panel **104**) along track **120-2** and allow a user or a device to place the transducer in and remove the transducer from transducer holder **109**. Spacing portion **127-3** may provide space for connection of the transducer to other devices, as described herein. For example, a wire of the transducer, which may extend from a bottom or second side of the transducer, may be connected to an oscilloscope or an evaluator device that enables determining whether the transducer under test is valid or not. For example, an invalid transducer may be a reverse poled transducer and/or has oppositely connected wires, and a valid transducer may be not reverse-poled and/or not have oppositely connected wires. For purposes of description, “reversed-poled” may mean where a positive wire is not connected to a positive-poled face of a piezoelectric component or a positive poled face of a piezoelectric component is not connected to a positive wire, for example.

(25) According to some exemplary embodiments, apparatus **100** may include spacers **130-1**, **130-2**, and **130-3** (also referred to collectively as spacers **130** and individually or generally as spacer **130**). Spacer **130** may be affixed to frame **103**, or spacer **130** may be a unitary element with frame **103**. Spacer **130-1** may have a vertical dimension that is substantially the same as spacing portion **127-1**. Additionally, spacer **130-2** may have a vertical dimension that is substantially the same as spacing portion **127-2** and spacer **130-3** may have a vertical dimension that is substantially the same as spacing portion **127-3**. Spacers **130** may also provide support for guide piece **107** and transducer holder **109** in a corresponding manner as upper lips **123** and lower lips **125**.

(26) According to some exemplary embodiments, frame **103** may include a recess **135** at which spacers **130** may be affixed or positioned. As illustrated, recess **135** may be adjacent to panels **137-1** and **137-2** (also referred to collectively as panels **137** and individually or generally as panel **137**) of frame **103**. Spacers **130** may be substantially flush with panels **137**. Additionally, as illustrated, gaps **139-1**, **139-2**, and **139-3** (also referred to collectively as gaps **139** and individually or generally as gaps **139**) may be configured between ends of spacers **130** and panels **137-1** and **139-2**. Gaps **139** may have widths that accommodate spacers **127** of panels **105**. As such, referring to FIGS. 1 and 2, spacers **130** and spacers **127** of panels **105** may form a squared-C configuration, which may be perpendicular to the squared-C configuration of tracks **120** of panels **105**.

Additionally, face plate **142** and face plates **143** may rest on panels **105**, which may help stabilize guide piece **107** and transducer holder **109**, as well as ensure that guide piece **107** and transducer holder **109** are properly seated relative to frame **103**, tracks **120**, spacers **130**, and so forth.

(27) FIG. **3** is a diagram illustrating an exemplary embodiment of a system **300**. For example, in addition to what has been previously illustrated and described, FIG. **3** shows a striker **145**, a controller **150**, a connector **155**, a validator **160**, and a transducer manager **170**. A wire **152** of a transducer is also illustrated, for purposes of description relating to system **300**. According to various exemplary embodiments, system **300** may include additional, different, and/or fewer elements. For example, an element (e.g., controller **150**, validator **160**, and/or transducer manager **170**) that provides an automated step(s) of a validation process may be omitted. Additionally, or alternatively, one or multiple steps performed by the element may be omitted without omitting the element in its entirety.

(28) As previously described, a striker, such as striker **145** may be used to hit the transducer under test to cause the transducer to produce a charge or voltage that may be measured and evaluated for purposes of validation. According to this example, striker **145** has a rod-like shape, but other configurations may be implemented. The material, weight, density, and other factors of relevance pertaining to striker **145** to provide sufficient force to produce the voltage/charge yet be non-destructive to the transducer is beyond the scope of this description. Optionally, striker **145** may include a handle portion **147**. The handle portion **147** can have a knurl or a method of preventing a slip of striker **145** from a hand on to the transducer in a manual validation configuration.

(29) Controller **150** may include a device that manages the operation of striker **145**. Controller **150** may include a mechanism that holds striker **145** in a position for release and subsequent striking of the transducer in a fully automated validation operation. Controller **150** may also include a mechanism for retrieving and placing striker **145** back in the position for a next test. Controller **150** may include logic for automating this process. For example, controller **150** may include timers that may relate to the releasing and retrieving striker **145**. Controller **150** may also be implemented to include a mechanism (e.g., magnetic, pneumatic, electro-magnetic, etc.) and/or other elements (e.g., a motor, a solenoid, etc.) to manage the operation of striker **145**. In this way, controller **150** may automate the use of striker **145** in the testing process. According to some exemplary embodiments, validator **160** may be connected to controller **150**. For example, validator **160** may control, communicate, trigger, and/or coordinate with controller **150** to provide an automation of a validation process, as described herein and further below. According to other exemplary embodiments, controller **150** may manage another type of device such as a commercial pulse generator, a signal generator, a pulse generator by a time-to-digital converter board in a flow meter, or another type of generator that may generate an impulse (e.g., an impulse voltage, an impulse force, etc.) that provides sufficient excitation of the piezoelectric transducer. The controller **150** may also instruct the next steps in an automated validation process.

(30) Wire **152** may be of a transducer **164**, which is illustrated and described in FIGS. **5A** and **5B**. For example, wire **152** may be implemented as a cable having a sheath surrounding +Ve and -Ve wires. The cable **152** may be of a three wire or a two wire configuration. A correctly poled piezoelectric transducer **164** may have a +Ve wire (e.g., indicated by color) connected to the positive poled-PZT material and a -Ve wire (e.g., indicated by a different color) connected to the negative-poled PZT material.

(31) Connector **155** may include a cable, wires, and/or include another type of interface that provides a connection between transducer **164** and validator **160**. For example, connector **155** may include a suitable plug, alligator clip, wire connector, or another type of interface that accommodates an input to validator **160** from wire **152** of transducer **164**. The cable **152** connector may be a three or two pin outlet with the sheath connected to the -Ve wire in the two pin configuration.

(32) Validator **160** may include a device that performs a validation process. Validator **160** may

include logic that measures the voltage, the signal, or the charge received via connector **155** from transducer **164**. Validator **160** may convert the received input from analog to digital. Validator **160** may include analytics that evaluates the measured voltage, signal, or charge and determines whether the transducer is validated or not. According to exemplary embodiment, the analytics may identify the first peak or wave of the measured signal at a zero crossing. A signal of a transducer that has a positive peak at the zero crossing may be deemed validated and a signal that has a first negative peak or is out of phase (e.g., 180 degrees, etc.) at the zero crossing may be deemed not validated. Validator **160** may store and apply a reference voltage to the signal to determine the first peak relative to an envelope of the signal over time. For example, there may be a positive reference voltage and/or a negative reference voltage for determining first peaks of correctly poled and/or incorrectly poled transducers **164**. According to some exemplary embodiments, validator **160** may be implemented as a computer that hosts software and/or other types of instructions, which when executed, is configured to perform the validation process or a portion thereof. According to some exemplary embodiments, validator **160** may include multiple devices, such as one device to measure the output of transducer **164** and another device to analyze or evaluate the measured output.

(33) Although not illustrated, validator **160** may be connected to, control, and/or trigger an operation of controller **150**. For example, the release of striker **145** and the retrieval of striker **145** (e.g., from transducer holder **109**) may be based on communication with validator **160**. By way of further example, after an output signal of transducer **164** is received by validator **160** or after the output signal of transducer **164** is analyzed, validator **160** may trigger or instruct controller **150** to retrieve striker **145** and transducer **164**. Removal of transducer **164** and/or replacement of the tested transducer **164** with another transducer **164** to be tested may subsequently occur, such as by an operator or a device (not illustrated), as previously described. Controller **150** may communicate or signal validator **160** to indicate the tested transducer **164** has been removed or apparatus **100** and/or indicate a new transducer **164** to be tested has been received. According to various exemplary embodiments, controller **150** may position transducer **164** in transducer holder **109** with or without communication and/or coordination with validator **160**. Controller **150** may communicate or signal to validator **160** that the new transducer **164** is in position (e.g., seated in transducer holder **109**) and ready for testing. Validator **160** may acknowledge and/or govern when striker **145** is released for striking the new transducer **164** with controller **150**.

(34) Transducer manager **170** may include a device that manages receipt and removal of transducer **164**. For example, transducer manager **170** may be implemented as a robotic arm device. Transducer manager **170** may include logic that places transducer **164** in transducer holder **109** and retrieves transducer **164**. According to some exemplary embodiments, the execution of placement and retrieval, by transducer manager **170**, may be coordinated with controller **150** and/or validator **160**. As illustrated in FIG. 3, for example, transducer manager **170** may be communicatively coupled to validator **160**. Although not illustrated, other connections may be implemented between controller **150**, validator **160**, and/or transducer manager **170** that may support various messages of communication pertaining to a validation process or step of the validation process.

(35) According to some exemplary embodiments, transducer manager **170** may also connect and/or disconnect wire **152** of transducer **164** to/from connector **155**, replace the transducer **164** with the next transducer once the test is complete from transducer holder **109** either by removing transducer holder **109** from panels **105** (e.g., via tracks **120**) or by replacing transducer **164** without removing transducer holder from panels **105**. In case of the controller **150** failure, striker **145** can be set in position to drop from a pre-determined height using transducer manager **170**. For example, the robotic arm may perform these tasks. According to other exemplary embodiments, these tasks may be performed, in whole or in part, by a person.

(36) FIG. 4A is a diagram that illustrates exemplary signals and their first peak characteristics, recorded in a flow meter as described herein. In a flow meter, assuming the transmitting transducer

is correctly connected/poled, the receive transducer can be validated. For example, first peak **405** of a signal associated with one transducer **164** may be indicative that transducer **164** is validated whereas first peak **410** of another signal associated with another transducer **164** may be indicative that the other transducer **164** is not validated (e.g., reverse-poled PZT of the transducer or its wires are oppositely connected). According to an exemplary embodiment, validator **160** may store and apply a reference voltage for analyzing a signal and determining a first peak, as described herein. (37) FIG. **4B** is another diagram illustrating exemplary signals and their first peak characteristics based on a mechanical force used to excite a piezoelectric transducer **164**. As illustrated, a first peak **415** of a first signal **414** may be indicative that transducer **164** is validated whereas a first peak **420** of a second signal **419** may be indicative that the other transducer **164** is not validated (e.g., reverse-poled PZT of the transducer or its wires are oppositely connected).

(38) As previously described, according to other embodiments, the validation process may be performed, at least in part, by a human. For example, an oscilloscope may be connected to transducer **164**, and the output voltage, signal, or charge may be received and displayed to a user. Based on the signature of the received input and a reference voltage, the user may determine whether transducer **164** is validated or not (e.g., correctly poled or not).

(39) According to an exemplary embodiment, transducer **164** is a piezoelectric transducer. For example, the piezoelectric may be implemented with various materials, such as $\text{Pb}(\text{ZrTi})\text{O}_{0.3}$, $\text{PbTiO}_{0.3}$, $\text{BaTiO}_{0.3}$, $\text{Bi}_{0.4}\text{Ti}_{0.3}\text{O}_{1.2}$, $\text{LiTiO}_{0.3}$, or another suitable piezoelectric material.

(40) FIGS. **5A** and **5B** are diagrams illustrating views of transducer holder **109** and transducer **164**. Referring to FIG. **5A**, as previously described, transducer holder **109** may include passthrough hole **110** that is configured to receive and secure the transducer under test. As further illustrated, within passthrough hole **110**, a seat **502** may be formed upon which transducer **164** may be seated and secured. For example, in the cross-sectional view of transducer holder **109** in FIG. **5B**, passthrough hole **110** may include a counterbored configuration having a first bore **510** and a second bore **515**. First bore **510** and second bore **515** may be of different diameters and form a seating surface, e.g., seat **502** with respect to a portion of a bottom or second surface of transducer **164**. For example, first bore **510** may be configured to accommodate the body of transducer **164**, and second bore **515** may be configured to accommodate wire **152** of transducer **164**. As such, second bore **515** may have a smaller diameter than first bore **510**. As further illustrated, transducer holder **109** may include a passthrough slot **505** that may accommodate wire **152** of transducer **164** when transducer **164** is received and removed or swapped in and out of transducer holder **109**. According to various exemplary embodiments, during a validation test, a top or first surface of transducer **164** may be above, flush with, or below, a top surface **520** of transducer holder **109**. According to some exemplary embodiments, seat **502** may include an O-ring (not illustrated) or another type of similar element. The O-ring may minimize (e.g., dampen) ringing of transducer **164** after being struck by striker **145** and assist in seating transducer **164** in transducer holder **109**. FIG. **6** is a diagram illustrating a top view of a portion of apparatus **100**.

(41) According to other exemplary embodiments, passthrough hole **110** may be of a different configuration. For example, depending on the shape, structure, contour, and/or other type of characteristic of transducer **164**, passthrough hole **110** may be of a configuration conducive to such characteristic(s) of transducer **164** and enable transducer **164** to be appropriately positioned in transducer holder **109** for the validation process. According to some exemplary embodiments, one or more dimensions of passthrough hole **110** may be configurable using bushings or another type of insertable interface to modify first bore **510** and/or second bore **515** to receive and secure transducers **164** of varying sizes and/or shapes.

(42) As previously mentioned, according to various exemplary implementations, transducers **164** may be received and removed or swapped in and out for testing by a human or use of a device.

(43) FIG. **7** is a diagram illustrating exemplary components of a device **700** that may be included in

one or more elements described herein. For example, controller **150** and/or validator **160** may include device **700** or a portion of (e.g., a component) device **700**. As illustrated in FIG. 7, device **700** includes a bus **705**, a processor **710**, a memory/storage **715** that stores software **720**, a communication interface **725**, an input **730**, and an output **735**. According to other embodiments, device **700** may include fewer components, additional components, different components, and/or a different arrangement of components than those illustrated in FIG. 7 and described herein.

(44) Bus **705** includes a path that permits communication among the components of device **700**. For example, bus **705** may include a system bus, an address bus, a data bus, and/or a control bus. Bus **705** may also include bus drivers, bus arbiters, bus interfaces, clocks, and so forth.

(45) Processor **710** includes one or multiple processors, microprocessors, data processors, co-processors, graphics processing units (GPUs), application specific integrated circuits (ASICs), controllers, programmable logic devices, chipsets, field-programmable gate arrays (FPGAs), application specific instruction-set processors (ASIPs), system-on-chips (SoCs), central processing units (CPUs) (e.g., one or multiple cores), microcontrollers, and/or some other type of component that interprets and/or executes instructions and/or data. Processor **710** may be implemented as hardware (e.g., a microprocessor, etc.), a combination of hardware and software (e.g., a SoC, an ASIC, etc.), may include one or multiple memories (e.g., cache, etc.), etc.

(46) Processor **710** may control the overall operation, or a portion of operation(s) performed by device **700**. Processor **710** may perform one or multiple operations based on an operating system and/or various applications or computer programs (e.g., software **720**). Processor **710** may access instructions from memory/storage **715**, from other components of device **700**, and/or from a source external to device **700** (e.g., a network, another device, etc.). Processor **710** may perform an operation and/or a process based on various techniques including, for example, multithreading, parallel processing, pipelining, interleaving, learning, model-based, etc.

(47) Memory/storage **715** includes one or multiple memories and/or one or multiple other types of storage mediums. For example, memory/storage **715** may include one or multiple types of memories, such as, a random access memory (RAM), a dynamic random access memory (DRAM), a static random access memory (SRAM), a cache, a read only memory (ROM), a programmable read only memory (PROM), an erasable PROM (EPROM), an electrically EPROM (EEPROM), a single in-line memory module (SIMM), a dual in-line memory module (DIMM), a flash memory (e.g., 2D, 3D, NOR, NAND, etc.), a solid state memory, and/or some other type of memory. Memory/storage **715** may include a hard disk (e.g., a magnetic disk, an optical disk, a magneto-optic disk, a solid-state disk, etc.), a Micro-Electromechanical System (MEMS)-based storage medium, and/or a nanotechnology-based storage medium.

(48) Memory/storage **715** may include drives for reading from and writing to the storage medium. Memory/storage **715** may be external to and/or removable from device **700**, such as, for example, a Universal Serial Bus (USB) memory stick, a dongle, a hard disk, mass storage, off-line storage, or some other type of storing medium (e.g., a compact disk (CD), a digital versatile disk (DVD), a Blu-Ray disk (BD), etc.). Memory/storage **715** may store data, software, and/or instructions related to the operation of device **700**.

(49) Software **720** includes an application or a program that provides a function and/or a process. As an example, with reference to controller **150**, software **720** may include an application that, when executed by processor **710**, provides a function and/or a process for the management of striker **145**, as described herein. According to another example, with reference to validator **160**, software **720** may include an application that, when executed by processor **710**, provides a function and/or a process pertaining to measurement and analysis of signals output from transducer **164**. Software **720** may also include firmware, middleware, microcode, hardware description language (HDL), and/or other form of instruction. Software **720** may also be virtualized. Software **720** may further include an operating system (OS) (e.g., Windows, Linux, Android, proprietary, etc.).

(50) Communication interface **725** permits device **700** to communicate with other devices,

networks, systems, and/or the like. Communication interface **725** includes one or multiple wireless interfaces and/or wired interfaces. For example, communication interface **725** may include one or multiple transmitters and receivers, or transceivers. Communication interface **725** may operate according to a protocol stack and/or a communication standard. Communication interface **725** may include various processing logic or circuitry (e.g., multiplexing/de-multiplexing, filtering, amplifying, converting, error correction, application programming interface (API), etc.).

(51) Input **730** permits an input into device **700**. For example, input **730** may include a keyboard, a mouse, a display, a touchscreen, a touchless screen, a button, a switch, an input port, speech recognition logic, and/or some other type of visual, auditory, tactile, etc., input component. Output **735** permits an output from device **700**. For example, output **735** may include a speaker, a display, a touchscreen, a touchless screen, a light, an output port, and/or some other type of visual, auditory, tactile, etc., output component.

(52) Device **700** may perform a process and/or a function, as described herein, in response to processor **710** executing software **720** stored by memory/storage **715**. By way of example, instructions may be read into memory/storage **715** from another memory/storage **715** (not shown) or read from another device (not shown) via communication interface **725**. The instructions stored by memory/storage **715** cause processor **710** to perform a function or a process described herein. Alternatively, for example, according to other implementations, device **700** performs a function or a process described herein based on the execution of hardware (processor **710**, etc.).

(53) FIG. **8** is a flow diagram illustrating an exemplary process **800** of an exemplary embodiment of the validation process. According to an exemplary embodiment, apparatus **100** or system **300** may perform, in whole or in part, steps of process **800**. According to an exemplary implementation, controller **150**, validator **160**, and/or transducer manager **170** may execute software to perform a step illustrated in FIG. **8** and described herein. Alternatively, a step illustrated in FIG. **8** and described herein, may be performed by execution of only hardware.

(54) Referring to FIG. **8**, in block **805**, a piezoelectric transducer is received. For example, transducer holder **109** may receive transducer **164** to be tested. Depending on the scenario, this may be part of a first placement process, in which transducer holder **109** does not have a transducer **164**. For example, this may occur when a first-to-be-tested transducer **164** is received. Alternatively, when multiple transducers **164** are to be tested, this may be part of a swap-in and swap-out process, in which a user and/or transducer manager **170** has retrieved a tested transducer **164** from transducer holder **109** after having struck transducer **164**, and a new transducer **164** is to be received. In this regard, according to various exemplary implementations, transducers **164** may be received (and removed) or swapped in and out for testing by a human or used by transducer manager **170**.

(55) According to an exemplary embodiment, a striking surface of striker **145** may be in a position that is in a space between guide piece **107** and transducer holder **109**. For example, as previously mentioned, spacing portion **127-2** may afford sufficient space between a bottom surface of guide piece **107** and a top surface of transducer holder **109** to allow transducers **164** to be received and removed or swapped in and out for testing. According to an exemplary embodiment, wire **152** of transducer **164** may be connected to validator **160** by a user and/or transducer manager **170**.

(56) In block **810**, the piezoelectric transducer is impulsively stimulated (e.g., struck) to cause an output of a signal. For example, with transducer **164** retained in transducer holder **109**, striker **145** may impact transducer **164** and cause an output signal to be generated by transducer **164**. According to various exemplary embodiments, depending on the extent of automation (e.g., fully or partially) or not (e.g., wholly performed by a user), this step may be initiated based on communication between elements of system **300**, such as validator **160**, transducer manager **170**, and/or controller **150**, as described herein. For example, the communication may include a message indicating that that transducer **164** is ready to be stimulated. Additionally, striker **145** may be moved from a position that placed or allowed transducer **164** to be seated in transducer holder **109**

to another position from which striker **145** may initiate hitting transducer **164**.

(57) In block **815**, the signal is measured. For example, validator **160** may receive the output signal generated by transducer **164** via connector **155**. Validator **160** may measure the received signal over a certain duration of time.

(58) In block **820**, one or more characteristics of the measured signal is evaluated. For example, validator **160** may analyze the measured signal based on a reference voltage, as described herein. For example, validator **160** may determine a first peak of the measured signal and its polarity (e.g., positive or negative), as described herein.

(59) In block **825**, an indication whether the polarity of the piezoelectric transducer is correct or not is provided. For example, validator **160** may determine whether transducer **164** is correctly poled/wire connected or not based on the first peak information. Validator **160** may provide a result of the test that indicates that transducer **164** is correctly poled/wired connected or not. For example, the result may be displayed, data may be written and stored, and/or other types of outputs may be provided to indicate the result. According to some implementations, an identifier of transducer **164** may be correlated to the result. According to some exemplary embodiments, validator **160** may communicate with other elements (e.g., transducer manager **170** and/or controller **150**) and/or a user in response to determining a result. The communication may trigger the removal of transducer **164**.

(60) In block **830**, the piezoelectric transducer is removed. For example, transducer manager **170** or the user may disconnect transducer **164** from validator **160**. Controller **150** may position transducer **164** for removal by transducer manager **170** or the user.

(61) FIG. **8** illustrates an exemplary process **800** of the validation process, however, according to other embodiments, process **800** may include additional operations, fewer operations, and/or different operations than those illustrated in FIG. **8** and described herein. Process **800** may be iteratively performed in a fully automated or semi-automated manner, for example.

(62) FIG. **9** is a flow diagram illustrating another exemplary process **900** of an exemplary embodiment of the validation process. According to an exemplary embodiment, apparatus **100** or system **300** may perform, in whole or in part, steps of process **900**. According to an exemplary implementation, controller **150**, validator **160**, and/or transducer manager **170** may execute software to perform a step illustrated in FIG. **9** and described herein. Alternatively, a step illustrated in FIG. **9** and described herein, may be performed by execution of only hardware.

(63) Referring to FIG. **9**, in block **905**, the piezoelectric transducer is stimulated to cause an output of a signal. For example, transducer **164** may be physically excited in a manner that causes transducer **164** to output a signal. According to some exemplary embodiments, transducer **164** may be assembled with other elements of a device like in a flow meter, where a transmitting transducer is connected to a generator (e.g., a pulse generator) and the receiving transducer is connected to measure the time-of-flight data that is converted to volume flow. According to other exemplary embodiments, transducer **164** may be an isolated element where it can be used to detect flaws in solids, levels in liquids, create water mist for therapeutics and provide medical imaging as an echo generator, for example.

(64) In block **910**, the signal is measured. For example, validator **160** may receive the output signal generated by transducer **164** via connector **155**. Validator **160** may measure the received signal over a certain duration of time.

(65) In block **915**, one or multiple characteristics of the measured signal is evaluated. For example, validator **160** may analyze the measured signal based on a reference voltage, as described herein. For example, validator **160** may determine a first peak of the measured signal and its polarity (e.g., positive or negative), as described herein.

(66) In block **920**, an indication whether the polarity of the piezoelectric transducer is correct or not is provided. For example, validator **160** may determine whether the transducer **164** is correctly poled/wire connected or not based on the first peak information. Validator **160** may provide a result

of the test that indicates that transducer **164** is correctly poled/wire connected or not. For example, the result may be displayed, data may be written and stored, and/or other types of outputs may be provided to indicate the result. According to some implementations, an identifier of transducer **164** may be correlated to the result. According to some exemplary embodiments, validator **160** may communicate with other elements (e.g., transducer manager **170** and/or controller **150**) and/or a user in response to determining a result.

(67) In block **925**, a response based on the indication is determined and/or executed. For example, when the polarity of transducer **164** is correct, validator **160** or another device may determine transducer **164** is validated for assembly or use in a device. For example, the device may be a water meter. Alternatively, for example, when the polarity of transducer **164** is incorrect, validator **160** or another device, which may receive the result of the determination, may determine whether to use transducer **164** or not. According to an exemplary implementation, even when the polarity of transducer **164** is incorrect, transducer **164** may be used. A reverse poled/connected transducer in a water meter will record a time-of-flight offset (e.g., illustrated in FIG. **4A**) that will affect the calculated flow volume. In that case for example, logic of the water meter may be modified in a manner that accounts for the reverse-poled transducer time-of-flight offset. This method holds good in any application where there is an offset in time-of-flight in a transmit-receive measurement.

(68) FIG. **9** illustrates an exemplary process **900** of the validation process, however, according to other embodiments, process **900** may include additional operations, fewer operations, and/or different operations than those illustrated in FIG. **9** and described herein. For example, process **900** may include a step similar to that described in block **805** of process **800**. Process **900** may be iteratively performed in a fully automated or semi-automated manner, for example.

(69) As set forth in this description and illustrated by the drawings, reference is made to “an exemplary embodiment,” “an embodiment,” “embodiments,” etc., which may include a particular feature, structure, or characteristic in connection with an embodiment(s). However, the use of the phrase or term “an embodiment,” “embodiments,” etc., in various places in the specification does not necessarily refer to all embodiments described, nor does it necessarily refer to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiment(s). The same applies to the term “implementation,” “implementations,” etc.

(70) The foregoing description of embodiments provides illustration, but is not intended to be exhaustive or to limit the embodiments to the precise form disclosed. Accordingly, modifications to the embodiments described herein may be possible. For example, various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The description and drawings are accordingly to be regarded as illustrative rather than restrictive.

(71) According to some exemplary embodiments, the antenna assembly detection service may be used in combination (e.g., parallel, in series) with other antenna detection approaches. For example, reference is made to pending provisional patent applications that describe antenna detection based on noise measurement and return loss (e.g., reflected power and forward power).

(72) The terms “a,” “an,” and “the” are intended to be interpreted to include one or more items. Further, the phrase “based on” is intended to be interpreted as “based, at least in part, on,” unless explicitly stated otherwise. The term “and/or” is intended to be interpreted to include any and all combinations of one or more of the associated items. The word “exemplary” is used herein to mean “serving as an example.” Any embodiment or implementation described as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or implementations.

(73) Embodiments described herein may be implemented in many different forms of software executed by hardware. For example, a process or a function may be implemented as “logic,” a “component,” or an “element.” The logic, the component, or the element, may include, for example, hardware (e.g., processor **710**, etc.), or a combination of hardware and software.

(74) Embodiments have been described without reference to the specific software code because the software code can be designed to implement the embodiments based on the description herein and commercially available software design environments and/or languages. For example, various types of programming languages including, for example, a compiled language, an interpreted language, a declarative language, or a procedural language may be implemented.

(75) Additionally, embodiments described herein may be implemented as a non-transitory computer-readable storage medium that stores data and/or information, such as instructions, program code, a data structure, a program module, an application, a script, or other known or conventional form suitable for use in a computing environment. The program code, instructions, application, etc., is readable and executable by a processor (e.g., processor **710**) of a device. A non-transitory storage medium includes one or more of the storage mediums described in relation to memory/storage **715**. The non-transitory computer-readable storage medium may be implemented in a centralized, distributed, or logical division that may include a single physical memory device or multiple physical memory devices spread across one or multiple network devices.

(76) Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another, the temporal order in which acts of a method are performed, the temporal order in which instructions executed by a device are performed, etc., but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

(77) The terms “about” and “approximately” shall generally mean an acceptable degree of error or variation for the quantity measured given the nature or precision of the measurements. Typical, exemplary degrees of error or variation are within 20 percent (%), preferably within 10%, and more preferably within 5% of a given value or range of values. Numerical quantities given in this description are approximate unless stated otherwise, meaning that the term “about” or “approximately” can be inferred when not expressly stated.

(78) No element, act, or instruction set forth in this description should be construed as critical or essential to the embodiments described herein unless explicitly indicated as such.

(79) All structural and functional equivalents to the elements of the various aspects set forth in this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. No claim element of a claim is to be interpreted under 35 U.S.C. § 112(f) unless the claim element expressly includes the phrase “means for” or “step for.”

Claims

1. A device comprising: a frame; a striker; a guide piece that includes a passthrough hole configured for the striker; a transducer holder that is configured to position a piezoelectric transducer for test, wherein the transducer holder, which is positioned beneath the guide piece, exposes a first side of the piezoelectric transducer and a second side of the piezoelectric transducer; and a first controller configured to: analyze an output signal of the piezoelectric transducer caused by the striker traveling via the passthrough hole and hitting the first side of the piezoelectric transducer; and determine, based on the analysis, whether a polarity of the piezoelectric transducer is correct or not.
2. The device of claim 1, wherein the transducer holder includes a counterbore configuration that includes a first bore that exposes the first side of the piezoelectric transducer and a second bore that exposes the second side of the piezoelectric transducer, wherein the first bore has a larger diameter than the second bore.
3. The device of claim 2, wherein the transducer holder includes a passthrough slot that is configured to receive a wire of the piezoelectric transducer, wherein the passthrough slot is

connected to the first bore and the second bore.

4. The device of claim 2, wherein the second bore exposes a wire of the piezoelectric transducer, and the first controller is configured to connect to the wire of the piezoelectric transducer.

5. The device of claim 1, wherein, when determining, the first controller is further configured to: determine a first peak of the output signal based on a reference voltage; and determine whether the first peak is a positive voltage or a negative voltage.

6. The device of claim 5, wherein the first controller is further configured to: determine that the polarity of the piezoelectric transducer is incorrect when the first peak is a negative voltage; and determine that the polarity of the piezoelectric transducer is correct when the first peak is a positive voltage.

7. The device of claim 1, further comprising: a second controller configured to: cause the striker to hit the first side of the piezoelectric transducer; and retrieve the striker after the first side of the piezoelectric transducer is hit.

8. The device of claim 1, wherein the frame includes a first track configured to receive the guide piece and a second track configured to receive the transducer holder.

9. A device comprising: a frame; a generator configured to generate an impulse sufficient to excite a piezoelectric transducer under test; a transducer holder that is configured to position the piezoelectric transducer, wherein the transducer holder exposes a first side of the piezoelectric transducer and a second side of the piezoelectric transducer; and a first controller configured to: analyze an output signal of the piezoelectric transducer caused by the generator; and determine, based on the analysis, whether a polarity of the piezoelectric transducer is correct or not.

10. The device of claim 9, wherein the transducer holder includes a counterbore configuration that includes a first bore that exposes the first side of the piezoelectric transducer and a second bore that exposes the second side of the piezoelectric transducer, wherein the first bore has a larger diameter than the second bore.

11. The device of claim 10, wherein the transducer holder includes a passthrough slot that is configured to receive a wire of the piezoelectric transducer, wherein the passthrough slot is connected to the first bore and the second bore.

12. The device of claim 9, wherein the impulse is an ultrasonic pulse.

13. The device of claim 9, wherein, when determining, the first controller is further configured to: determine a first peak of the output signal based on a reference voltage; and determine whether the first peak is a positive voltage or a negative voltage.

14. The device of claim 9, wherein the frame comprises: a first panel; and a second panel, wherein the first panel and the second panel are parallel to each other and are perpendicular to a base panel, and wherein the first panel and the second panel each includes a track that is configured to receive the transducer holder.

15. The device of claim 14, wherein the tracks of the first panel and the second panel allow the transducer holder to slide towards and slide away from the base panel.

16. A method comprising: causing a piezoelectric transducer to emit a signal; evaluating the signal based on a reference voltage; determining, based on the evaluating, whether the piezoelectric transducer is reverse-poled or not; and outputting a result of the determining, wherein the result indicates that the piezoelectric transducer is reverse-poled or that the piezoelectric transducer is correctly poled.

17. The method of claim 16, further comprising: determining to modify a first device that includes the piezoelectric transducer or to modify a second device that the piezoelectric transducer is to be included when the result indicates that a polarity of the piezoelectric transducer is not correct.

18. The method of claim 17, wherein the first device or the second device is a water meter.

19. The method of claim 17, wherein software or firmware of the first device or the second device is modified.

20. The method of claim 16, wherein the evaluating comprises: determining a first peak of the

signal based on the reference voltage; and determining whether the first peak is a positive voltage or a negative voltage.

21. A non-transitory computer-readable storage medium storing instructions executable by a processor of a device, which when executed cause the processor to: cause a piezoelectric transducer to emit a signal; evaluate the signal based on a reference voltage; determine, based on the evaluation, whether a polarity of the piezoelectric transducer is correct or not; and output a result of the determination, wherein the result indicates that the polarity of the piezoelectric transducer is correct or not.
